Mini EDA Report: Employee Dataset Analysis

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Tools used: Python, Pandas, Matplotlib, Seaborn (Jupyter Notebook in VS Code)
Task: Task 14, Exploratory Data Analysis (cleaning, summary statistics, visualizations)

1. Introduction

Exploratory Data Analysis (EDA) is the process of examining a dataset to understand its structure, fix data quality problems, and find patterns before any deeper analysis. This report presents a mini EDA of a company's active employee dataset, using Pandas for data handling and Matplotlib/Seaborn for visualization.

2. Dataset Description

The dataset contains records of **800 employees** and **14 columns**, loaded from Active_Employees.csv.

Type	Columns
Identifier	Employee_ID, Employee_Name
Categorical	Gender, Department, Job_Title, City, State, Employment_Status, Grade
Numeric	Age, Salary, Performance_Score, Bonus
Date	Joining_Date

The data covers 7 departments (Sales, HR, IT, Operations, Marketing, Finance, Customer Support), 6 states, and 4 performance grades (A to D).

##MINI EDA: PALMER PENGUINS

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
```

✓ 8.9s

##1.LOAD DATA

```
df = pd.read_csv(r"C:\Users\khond\OneDrive\Desktop\Python Internship Task\Active_Employees.csv", encoding="latin1")
print("Shape:", df.shape)
display(df.head())
```

[3] ✓ 0.0s Python

```
Shape: (800, 14)
```

	Employee_ID	Employee_Name	Gender	Age	Department	Job_Title	City	State	Joining_Date	Salary	Performance_Score	Bonus	Employment_Status	Grade
0	10001	Pooja Deshmukh	Female	27	Sales	Sales Executive	Pune	Maharashtra	2024-10-17 00:00:00	45000	97	6167.77	Active	D
1	10010	Kavya Deshmukh	Female	53	HR	HR Executive	Ahmedabad	Gujarat	2023-10-08 00:00:00	55000	55	6638.16	Active	B
2	10013	Neha More	Female	40	IT	Software Developer	Aurangabad	Maharashtra	2025-09-20 00:00:00	93000	90	8455.52	Active	B
3	10016	Meera Kadam	Male	54	Operations	Operations Executive	Mumbai	Maharashtra	2025-11-22 00:00:00	60000	88	4846.59	Active	B
4	10017	Karan More	Male	28	HR	HR Executive	Aurangabad	Maharashtra	2022-05-23 00:00:00	53000	64	1997.20	Active	B

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3. Initial Data Inspection

The following checks were performed on the raw data:

- df.shape, df.info() and df.describe() to understand size, data types and ranges
- df.isnull().sum() to check missing values
- df.duplicated().sum() to check duplicate records
- unique() on categorical columns to check for inconsistent spellings

Findings: There were no missing values and no duplicate rows. Numeric ranges were realistic (Age 21 to 55, Salary 25,000 to 95,000, Performance Score 50 to 100). The Joining_Date column was stored as text instead of a date type.

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Employee_ID        800 non-null    int64  
 1   Employee_Name      800 non-null    str    
 2   Gender             800 non-null    str    
 3   Age                800 non-null    int64  
 4   Department         800 non-null    str    
 5   Job_Title          800 non-null    str    
 6   City               800 non-null    str    
 7   State              800 non-null    str    
 8   Joining_Date       800 non-null    str    
 9   Salary             800 non-null    int64  
 10  Performance_Score  800 non-null    int64  
 11  Bonus              800 non-null    float64
 12  Employment_Status  800 non-null    str    
 13  Grade              800 non-null    str    
dtypes: float64(1), int64(4), str(9)
memory usage: 87.6 KB
```

```python
print("Final shape:", df_clean.shape)
print("Missing values:", df_clean.isnull().sum().sum())
df_clean.dtypes
```

```
Final shape: (800, 15)
Missing values: 0

Employee_ID                    int64
Employee_Name                    str
Gender                           str
Age                            int64
Department                       str
Job_Title                        str
City                             str
State                            str
Joining_Date         datetime64[us]
Salary                         int64
Performance_Score              int64
Bonus                        float64
Employment_Status                str
Grade                            str
Joining_Year                   int32
dtype: object
```

```
df.describe()
```

	Employee_ID	Age	Salary	Performance_Score	Bonus
count	800.000000	800.000000	800.000000	800.000000	800.000000
mean	11248.571250	37.693750	54063.750000	73.925000	4885.632725
std	707.803652	10.384713	15399.468025	14.447155	2277.601571
min	10001.000000	21.000000	25000.000000	50.000000	1012.990000
25%	10684.500000	28.000000	43000.000000	61.000000	3168.587500
50%	11276.000000	37.000000	53000.000000	73.000000	4561.590000
75%	11834.000000	47.000000	64000.000000	86.000000	6313.610000
max	12500.000000	55.000000	95000.000000	100.000000	13599.890000

4. Data Cleaning

Step	Action	Reason
1	Created a copy of the data (df_clean)	To keep the original data safe
2	Converted Joining_Date to datetime and created a Joining_Year column	Enables date-based analysis
3	Removed leading/trailing spaces in text columns	Prevents the same value being counted as different categories
4	Checked and removed duplicates (by full row and by Employee_ID)	0 duplicates found, so no rows were removed
5	Checked categorical values	Values were consistent (Male/Female, A/B/C/D)
6	Checked outliers using the IQR method	Age, Salary, Performance_Score: 0 outliers. Bonus: 8 outliers
7	Kept the Bonus outliers	They are genuine high bonus values, not data errors

Missing values: none were found, so no imputation was needed.
Note: Employment_Status contains only “Active”, so it carries no analytical value and was not used in the charts.

Final cleaned dataset: 800 rows and 15 columns (including the new Joining_Year), with 0 missing values.

```python
df_clean["Joining_Date"] = pd.to_datetime(df_clean["Joining_Date"])
df_clean["Joining_Year"] = df_clean["Joining_Date"].dt.year
print(df_clean["Joining_Date"].dtype)
```

```
datetime64[us]
```

```python
before = len(df_clean)
df_clean = df_clean.drop_duplicates()
df_clean = df_clean.drop_duplicates(subset="Employee_ID")
print("Removed:", before - len(df_clean), "duplicate rows")
```

```
Removed: 0 duplicate rows
```

```python
num_cols = ["Age", "Salary", "Performance_Score", "Bonus"]
for col in num_cols:
    Q1, Q3 = df_clean[col].quantile([0.25, 0.75])
    IQR = Q3 - Q1
    n = ((df_clean[col] < Q1 - 1.5*IQR) | (df_clean[col] > Q3 + 1.5*IQR)).sum()
    print(col, ":", n, "outliers")
```

```
Age : 0 outliers
Salary : 0 outliers
Performance_Score : 0 outliers
Bonus : 8 outliers
```

```python
print("Final shape:", df_clean.shape)
print("Missing values:", df_clean.isnull().sum().sum())
df_clean.dtypes
```

```
Final shape: (800, 15)
Missing values: 0

Employee_ID                  int64
Employee_Name                  str
Gender                         str
Age                          int64
Department                     str
Job_Title                      str
City                           str
State                          str
Joining_Date        datetime64[us]
Salary                       int64
Performance_Score            int64
Bonus                      float64
Employment_Status              str
Grade                          str
Joining_Year                 int32
dtype: object
```

5. Summary Statistics

Metric	Mean	Min	Max
Age	37.69 years	21	55
Salary	54,064	25,000	95,000
Performance Score	73.9	50	100
Bonus	4,886	1,013	13,600

Grouped analysis:

- **Marketing** has the most employees (127).
- Average salary by department: **IT is highest (71,858)** and **Customer Support is lowest (41,231)**.
- Average salary by gender: Female 54,403 and Male 53,690.
- Average salary by grade: A 52,412, B 53,335, C 57,503, D 53,247.

6. Visualizations

Chart 1: Distribution of Salary (Histogram). Shows how salaries are spread across the 25,000 to 95,000 range. **[Add one line: where most employees fall.]**

```python
plt.figure(figsize=(8, 5))
sns.histplot(df_clean["Salary"], kde=True, color="steelblue")
plt.title("Distribution of Salary")
plt.xlabel("Salary")
plt.show()
```

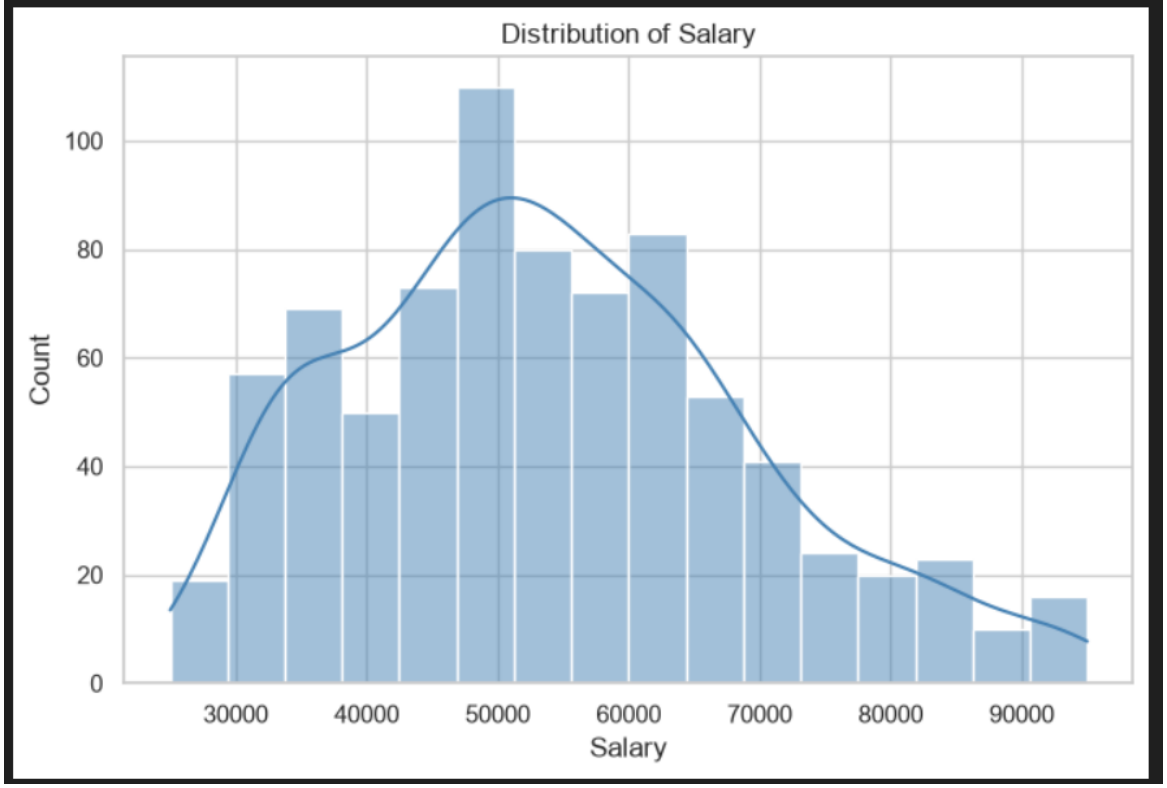


Chart 2: Employees by Department (Countplot). Compares headcount across the 7 departments. Marketing has the highest headcount.

```
##Chart 2: Department pramane employees (countplot)



plt.figure(figsize=(9, 5))
order = df_clean["Department"].value_counts().index
sns.countplot(data=df_clean, x="Department", order=order, palette="viridis")
plt.title("Employees by Department")
plt.xticks(rotation=30)
plt.show()
```

```
C:\Users\khond\AppData\Local\Temp\ipykernel_804\1063783086.py:3: FutureWarning:
```

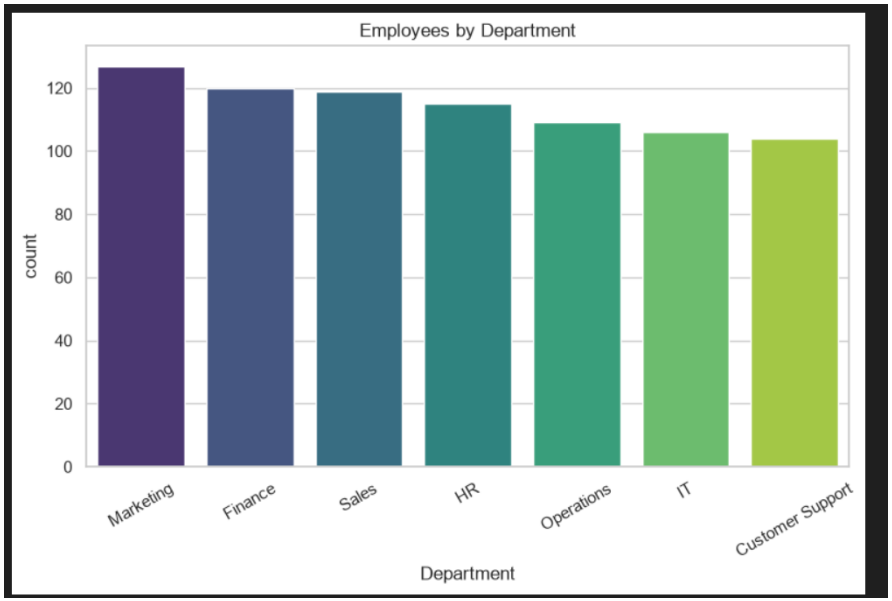


Chart 3: Salary by Department (Boxplot). Shows salary spread and median for each department. IT has the highest salaries and Customer Support the lowest.

```
##Chart 3: Department pramane Salary (boxplot)


plt.figure(figsize=(10, 6))
sns.boxplot(data=df_clean, x="Department", y="Salary", palette="Set2")
plt.title("Salary by Department")
plt.xticks(rotation=30)
plt.show()
```

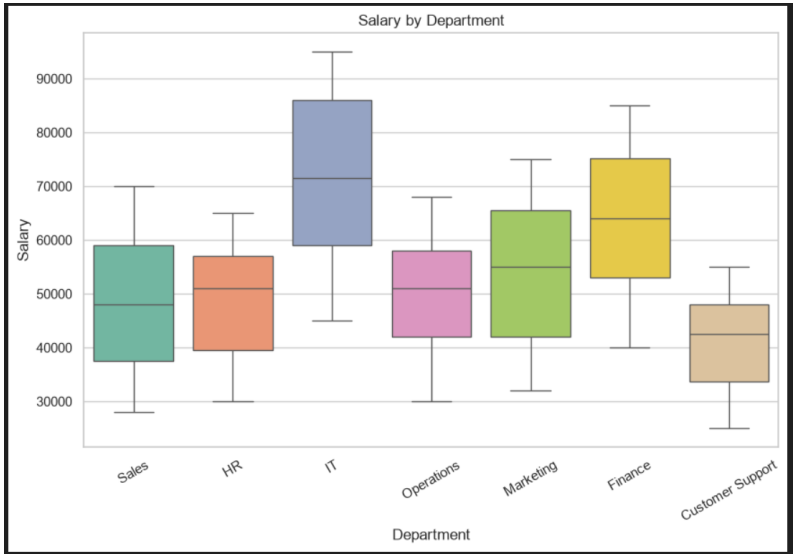
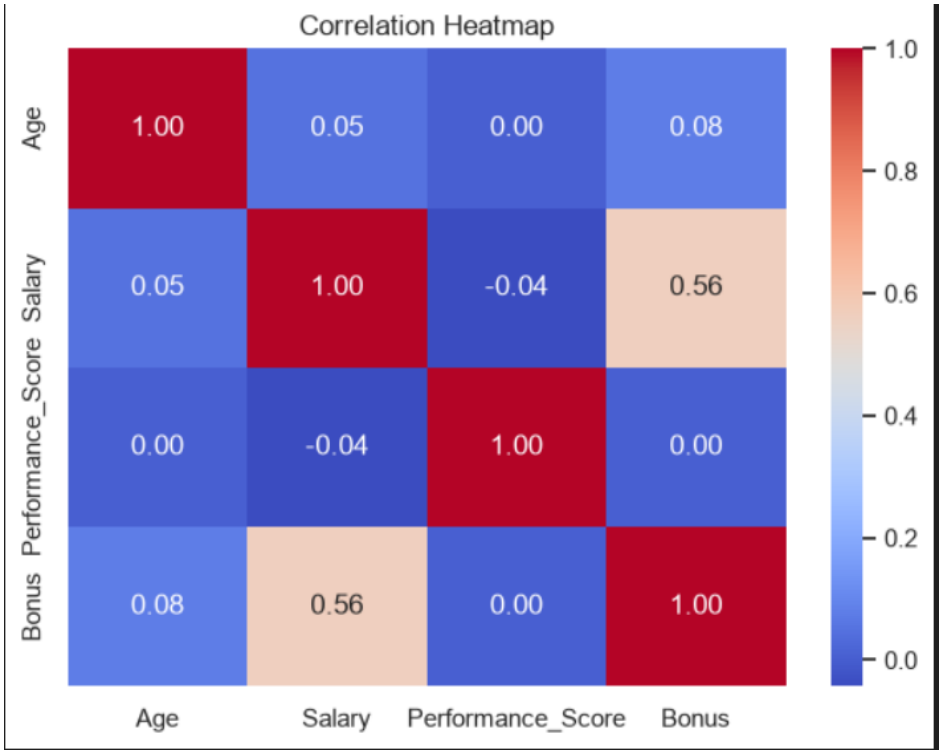


Chart 4: Correlation Heatmap. Shows relationships between Age, Salary, Performance Score and Bonus. The strongest relationship is between **Salary and Bonus (0.56)**, a moderate positive correlation.

```
##Chart 4: Correlation heatmap



plt.figure(figsize=(7, 5))
sns.heatmap(df_clean[["Age", "Salary", "Performance_Score", "Bonus"]].corr(),
            annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



7. Key Insights

1. **Department is the main driver of salary.** IT earns about 30,000 more on average than Customer Support.
2. **Gender pay difference is very small** (about 700 between averages), so no major gender-based gap appears in this data.
3. **Grade does not clearly relate to salary.** Grade C has the highest average salary, and Grade A the lowest, which is unusual and worth a closer look.
4. **Salary and Bonus are moderately correlated (0.56)**, so higher-paid employees tend to receive higher bonuses.
5. **2024 was the year with the most new joiners.**
6. **Data quality was good.** There were no missing values or duplicates.

8. Conclusion

This mini EDA examined 800 employee records. After checking and cleaning the data, summary statistics and four visualizations were used to understand the workforce. The analysis shows that salary depends mainly on department rather than gender or grade, and that salary and bonus are moderately related. These findings can help the company review its salary structure, plan hiring, and study how grades are assigned.